

A Differentiable Physics-Prior Residual 3D U-Net for Control-Point-Level Photon Dose Prediction on CT (DoseRAD2026 Photon-CT Task)

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Abstract. We predict the Monte-Carlo (MC) dose of every photon MLC control point with an analytical, differentiable physics prior followed by a residual 3D U-Net. Six channels (density, radiological depth, primary fluence, distance-to-CAX, source distance, and a naive analytical dose surrogate) are computed per control point from the CT density and the beam descriptor; a 3D U-Net (dilated bottleneck, Softplus head) learns only the residual to MC. Because the aperture enters as a fluence *input*, the model is robust to the benchmark’s deliberate shift from random training apertures to clinical VMAT test apertures. In ablation, the analytical prior contributes $+1.9\gamma(1\%/1\text{ mm})$ and hard-region loss weighting $+2.8$, stacking to $+4.0$; multi-modal joint training across real- and synthetic-CT inputs adds a further $+2.8$ (cross-validated, every fold). Under 5-fold cross-validation on the 75 public patients the model reaches plan $\gamma 1\%/1\text{ mm}$ of 93.4 ± 1.0 (multi-modal variant 96.2 ± 1.1). The final submission is trained on all 75 patients and deployed as a knowledge-distilled base-32 student ($2.1\times$ faster, *above* the teacher’s held-out gamma) with aperture-bbox cropping, GPU-resident channel construction, compiled inference and cutoff-zeroed output: on the hidden preliminary test set it scored $\gamma = 95.33$, per-beam MAE 0.0107, IDD 0.0072, stratified MAE 0.0049, DVH 0.30 at 80.3 s per job, well within the 181 s gate.

Keywords: photon dose prediction · differentiable physics · residual learning · knowledge distillation · VMAT

1 Introduction

Fast, accurate photon dose calculation underpins online adaptive radiotherapy: a plan must be recomputed on the anatomy of the day within minutes, at Monte-Carlo (MC)-level accuracy. Deep dose engines approach this [1,2], but most treat transport as an implicit target the network must memorize end-to-end. We instead factor the problem: an analytical, *differentiable* physics operator $\mathcal{C}$ maps

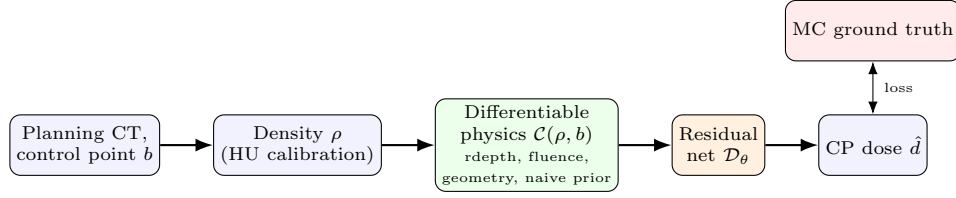


Fig. 1. Photon-CT pipeline: per control point, differentiable physics channels from the CT density; a 3D U-Net corrects the analytical prior toward MC.

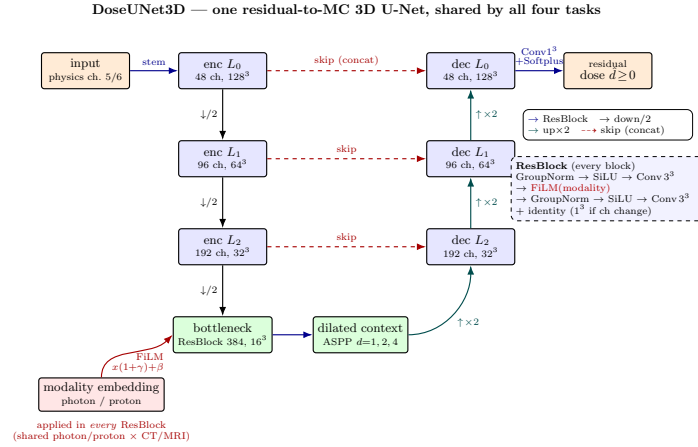


Fig. 2. The dose network $\mathcal{D}_\theta$ (DoseUNet3D): residual-to-MC 3D U-Net (channels 48/96/192/384; the deployed distilled student uses base 32), strided-conv down / transpose-conv up, dilated-context (ASPP, $d=1, 2, 4$) bottleneck for long-range lateral scatter, Softplus head (non-negative dose), FiLM modality conditioning in every ResBlock.

the CT density and the control-point descriptor to dose-shaped input channels, and a light 3D U-Net $\mathcal{D}_\theta$ learns only the residual between this prior and MC. The factorization has three measured consequences. It reduces learning to a smooth, bounded correction, so a light network trains stably; it makes the density-dependence explicit (the same operator powers our MRI-task synthesis training); and it is what makes the model robust to the benchmark’s deliberate aperture shift; training beams are *random* MLC apertures while test beams are *real clinical VMAT* apertures, and because the aperture is consumed as a fluence input rather than memorized, this shift is absorbed by construction. This report details the Photon-CT system; the same framework (shared architecture, different channels) serves our submissions to all four DoseRAD2026 tasks.

2 Methods

2.1 Data

We use only the official DoseRAD2026 training set [6] (75 patients: 39 thorax, 36 abdomen), built on the SynthRAD2025 cohort [7]: per patient a planning CT, beam parameters, and per-control-point Monte-Carlo dose-to-medium in absolute Gy (Geant4 full particle transport; the VMAT plans themselves are generated with matRad [4]) on a $2\times 2\times 2$ mm grid; 540 control points per patient. Training apertures are random; test apertures are real clinical VMAT (a deliberate distribution shift, Sec. 1). No external data are used for this task. For development we ran 5-fold cross-validation (CV) over all 75 patients (every patient held out exactly once); **the final submitted model is retrained on all 75 patients**, with the challenge leaderboard as the held-out validation signal.

2.2 Model

$\hat{d} = \mathcal{D}_\theta(\mathcal{C}(\rho, b))$, with ρ from the official piecewise-linear HU $\rightarrow$ density calibration. The operator $\mathcal{C}$ assembles **six channels** per control point, all implemented as differentiable tensor operations: (1) *density* ρ ; (2) *radiological depth*, the line integral $\int \rho d\ell$ from source to voxel, ray-marched along divergent rays with trilinear sampling; the water-equivalent depth that drives primary attenuation and build-up; (3) *primary fluence*: the MLC/jaw aperture of the control point projected through the divergent beam geometry onto each voxel and attenuated by $\exp(-\mu r \text{depth})$ with a 6 MV effective coefficient; (4) *distance to the central axis* (penumbra and off-axis softening); (5) *source distance* (inverse-square fall-off); (6) a *naive analytical dose surrogate* (fluence $\times$ inverse-square $\times$ build-up), a smooth, physically grounded starting estimate the network corrects. A per-ray *skin-entry* rule governs the depth and prior: along each ray, depth accumulates only from the first voxel with $\rho > 0.05 \text{ g/cm}^3$, and the prior is masked strictly upstream of that crossing: no dose before the body, nothing suppressed inside it (internal air cavities preserved). $\mathcal{D}_\theta$ is DoseUNet3D (Fig. 2): base 48 (teacher), four levels, ASPP bottleneck, Softplus head, FiLM conditioning; the *deployed* network is a base-32 student distilled from it (Sec. 2.3). Figure 3 visualizes the full residual-learning chain of the deployed model.

2.3 Training

Loss: high-dose-weighted L_1 to MC (high-dose fraction 0.1, weight 10) plus bounded penumbra/interface/lung terms; AdamW (lr 10^{-4} scratch, 2×10^{-5} finetune), EMA, gradient clipping; patches 128^3 , batch 2–4. The base-48 teacher trains $\sim 120\text{k}$ steps plus finetune. Two further stages define the final submission. **Multi-modal joint training:** starting from the trained photon-MRI system, the shared dose network continues training on a 50/50 interleave of real-CT cached-channel batches and MRI end-to-end steps; the resulting single network

Photon-CT (skin-entry) - 1ABB006 CP 0_134 (abdomen) | prior->GT corr 0.898 pred rel-MAE(>5%) 2.6%

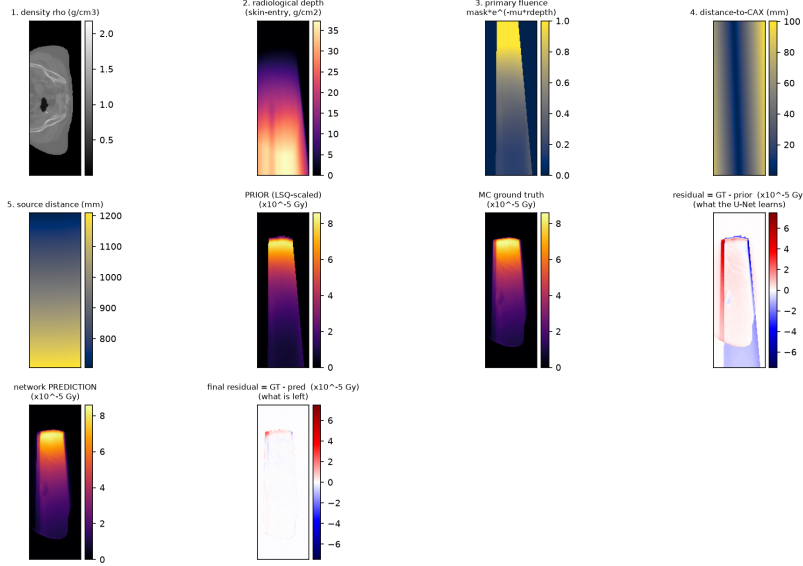


Fig. 3. Residual-learning chain of the deployed skin-entry model (abdomen, 1ABB006, peak-dose control point): input channels (density, radiological depth, primary fluence, distance-to-CAX, source distance) $\rightarrow$ LSQ-scaled naive prior $\rightarrow$ MC ground truth $\rightarrow$ residual (GT–prior) $\rightarrow$ network prediction $\rightarrow$ final residual (GT–pred). Both residual panels share one symmetric window: the shrinkage is the network’s contribution.

is a *better* CT engine than the dedicated CT model (Sec. 3.3). **Knowledge distillation (deployment):** the base-48 network is distilled into a base-32 student (7.6M vs. 17.1M parameters, $2.1\times$ faster forward) with a two-stage D-recipe: 200k steps of $L_1(\text{GT}) + 0.5 L_1(\text{teacher})$, then a 40k GT-only finetune. The student finished *above* the teacher’s held-out gamma (+0.33), so the speed is free. Reproducibility: fixed seeds, YAML-configured runs, EMA checkpoints; code and weights to be open-sourced (Sec. 6).

2.4 Evaluation

Internal: plan-level local γ [5] (pymedphys, 1%/1 mm, $\geq 10\%$ Rx cutoff, matching the official metric; verified byte-identical against the vendored official evaluation code), per-beam masked MAE, official IDD and stratified-MAE reimplementations, and a frozen 16-patient held-out cohort gating every deployment change (rule: *a change ships only if it loses neither accuracy nor speed*). A pitfall we surfaced and corrected: a photon control point has broad scatter tails ($\sim 4\%$ of plan energy outside margin-8 crops), and scoring against equally-cropped ground truth hides the loss entirely (96.2 vs. 86.4 against the full-grid GT on one patient); all crop margins were therefore selected on *full-grid* metrics. Runtime:

Table 1. Photon-CT results. CV: 5-fold over the 75 public patients. Hidden test: official metrics of the final full-data submission on the DoseRAD2026 preliminary test set (August 2026). The runtime gate is 181 s/job.

Metric	Value
CV plan $\gamma 1\%/1\text{ mm}$ (5-fold)	93.4 ± 1.0 (multi-modal variant 96.2 ± 1.1)
Hidden-test per-beam masked MAE	0.0107 ± 0.0014
Hidden-test IDD curve distance	0.0072 ± 0.0040
Hidden-test stratified plan MAE	0.0049 ± 0.0016
Hidden-test plan $\gamma 1\%/1\text{ mm}$	95.33 ± 1.89
Hidden-test DVH clinical score	0.30 ± 0.14
Hidden-test runtime per job	80.3 s (0.12 s/CP; distilled student ~ 0.09 s/CP)

the official score is $T = t_{\text{fix}} + N \cdot t_{\text{dose map}}$, gate 181 s/job, double-weighted in the ranking; we profile at job scale under the platform memory budget.

3 Results

3.1 Accuracy: cross-validated and hidden-test

Table 1 summarizes accuracy and the official hidden-test metrics. Per-fold CV $\gamma 1/1$: 94.0/92.4/94.4/94.2/92.1 (mean 93.4 ± 1.0); the multi-modal variant reaches 96.2 ± 1.1 , a +2.8 gain holding on every fold. On the fixed development fold ($n = 16$): $\gamma 1/1 = 95.7 \pm 3.8$ (abdomen 97.4, lung 94.0), $\gamma 3/3 = 99.8$; per-beam masked MAE $1.19 \pm 0.24\%$ of the beam maximum. For reference, the challenge’s analytical baseline (pyRadPlan) reaches only $\gamma 1/1 \approx 26.9$ on this metric; the residual network’s correction of the analytical prior is essential.

3.2 Ablations

The two Photon-CT-specific levers stack (single held-out fold, $n = 16$; Table 2): the naive analytical prior contributes +1.9 γ and hard-region loss weighting +2.8, together +4.0 over a plain 5-channel U-Net with uniform L_1 .

Table 2. Photon-CT ablation (plan $\gamma 1\%/1\text{ mm}$, $n = 16$): analytical prior $\times$ hard-region loss weighting.

	uniform L_1	hard-region weighting
no prior (5-ch)	91.7	94.5
+ naive prior (6-ch)	93.6	95.7

3.3 Multi-modal training exploits the photon residual’s head-room

A cross-validated *real-CT ceiling* experiment first located spare capacity: fed the real CT, the jointly trained photon-*MRI* dose network outperformed the dedicated CT network by +2.4 points (95.8 ± 1.3 vs. 93.4 ± 1.0), verified through the identical cached-channel pipeline (weights are the only difference). Testing the implied strategy directly, multi-modal joint training of one network over real- and synthetic-CT inputs raised CV Photon-CT to 96.2 ± 1.1 (+2.8, on every fold) with the MRI task unchanged. The mechanism is coherent with the residual design: the photon analytical prior is deliberately crude (it omits scatter, heterogeneity, penumbra), so the residual is large and structured, and exposure to a broader distribution of density-to-dose mappings acts as a strong, dose-relevant input augmentation. The identical experiment on the proton tasks; where the pencil-beam prior already supplies $\sim 95\%$ of the dose; is flat, a saturation control that locates the exploitable head-room precisely in the residual to a *weak* prior.

3.4 The crop margin is an accuracy–runtime knob, and cropped scoring hides its cost

We quantified a methodological pitfall that affects any cropped evaluation. A photon control point has broad scatter tails: $\sim 4\%$ of the plan’s energy falls outside margin-8 crops, concentrated in the 10–20%-of-max dose band where a *local* 1% gamma criterion has almost no tolerance. Scoring a margin-8 model against equally-cropped cached ground truth hid this entirely; 96.2 apparent vs. 86.4 true against the raw full-grid ground truth on one patient, so every margin decision was re-made on full-grid metrics. Widening the margin needs no retraining (the network is fully convolutional over analytic channels), and margin trades directly against runtime: margin 16 costs +0.01–0.02 s/CP over margin 8 but preserves the tail dose; margins 8 and 12 lost gamma and IDD on the board; a margin-8 *finetune* (network retrained at the deployed margin) did not recover the gap, showing the loss is physical tail truncation, not train–deploy mismatch. Margin 16 with cutoff-zeroed output is the deployed optimum. A related official-metric detail: the challenge’s ground truth zeroes dose below the per-beam cutoff; matching it by zeroing our sub-cutoff outputs was worth +3 γ on the photon board (for protons, quantising rather than zeroing was better), an example of why we vendored and byte-verified the official evaluation code.

3.5 Distillation: the student overtakes the teacher

The deployed network is a base-32 student (7.6M parameters) of the base-48 teacher (17.1M). Single-stage distillation converges slowly (fold-0 plan γ vs. steps: 77.7 at 40k, 86.9 at 80k, 93.2 at 160k, 94.0 at 200k = teacher parity 94.0), and the full D-recipe adds a 40k GT-only finetune, after which the full-data student *exceeded* the teacher’s held-out gamma by +0.33 at $2.1\times$ speed. Two scaling observations: a base-24 student ($2.6\times$) fell measurably short in probes, and cross-domain reuse fails; the CT-trained student loses 4.1 γ when stitched behind an

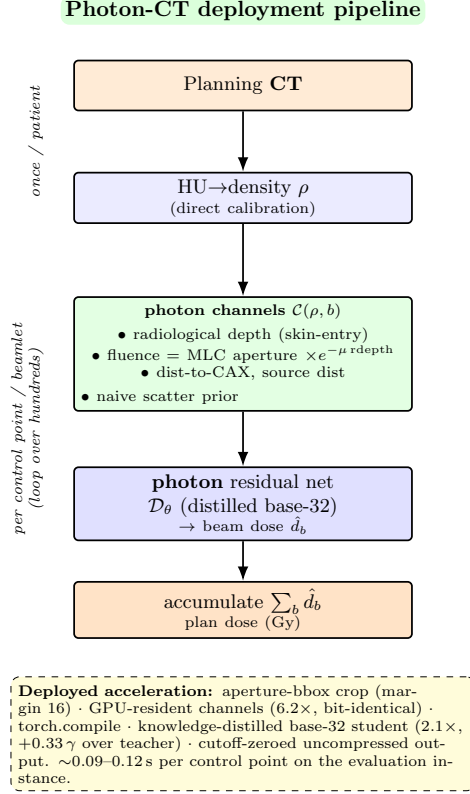


Fig. 4. The deployed Photon-CT container: per patient, the density stage runs once; the differentiable-channels $\rightarrow$ residual-net loop runs per control point/beamlet. The callout lists the deployed acceleration measures and their measured effects.

MRI front-end whose synthetic-density channels it never saw, recoverable only by finetuning on synthesis-domain channels. Distillation is thus domain-specific but, within domain, strictly dominant at this task size.

3.6 Runtime engineering

Runtime is double-weighted, so deployment was treated as a research target under the no-accuracy-loss gate. Table 3 lists the levers. Per control point, an aperture-derived bounding box (margin 16 voxels, selected on full-grid metrics) crops the compute domain; all six channels are built GPU-resident (6.2 $\times$ over the CPU builder, bit-identical); the network is compiled (`torch.compile` pays off here: one grid per patient, 540 same-shape control points amortize the one-time cost); sub-cutoff outputs are zeroed and written uncompressed (photon jobs are compute-bound; compression only adds CPU). The deployed distilled stu-

Table 3. Photon-CT acceleration levers (all accuracy-gated on a frozen held-out cohort; measured, not estimated).

Lever	Effect	Accuracy impact
Aperture-bbox crop, margin 16	−30 s/job vs. margin 24	γ preserved (full-grid gated)
GPU-resident channel builder	6.2 $\times$ channel build	bit-identical
<code>torch.compile</code> (dose net)	$\sim 1.5\times$ net	exact
Distilled base-32 student (D-recipe)	2.1 $\times$ net forward	+0.33 γ over teacher
Zero-cutoff output, no compression	write path off critical path	+3 γ (cutoff match)
<i>Rejected:</i> CP batching / async emission / fp16	$\leq 1.09\times$ or negative	—

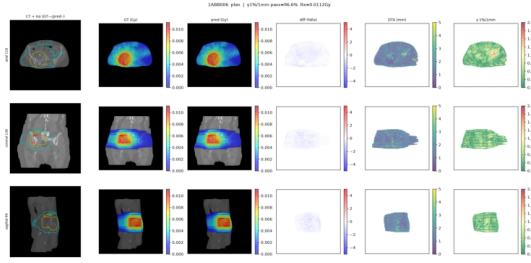
dent runs at ~ 35 ms/CP on our development GPU and ~ 0.09 – 0.12 s/CP on the evaluation instance. A negative finding worth reporting: for this small distilled network, batch-1 inference already saturates the GPU; control-point batching (1.01 – $1.09\times$), asynchronous delayed emission ($0.90\times$), fp16 weights ($1.00\times$) and autotuned compilation (—, platform t_{fix} pays the longer autotune) all failed our gate; once channel construction and the write path are off the critical path, the convolution forward itself is the deployment floor.

4 Discussion

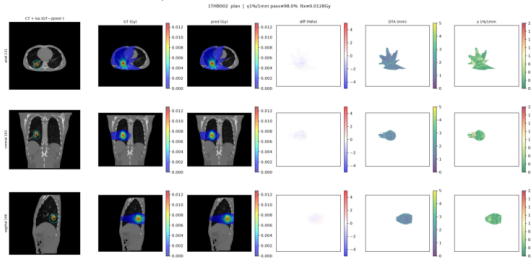
The explicit prior-plus-residual factorization, rather than raw capacity, drives this system’s accuracy, robustness and speed: the ablation shows prior and loss-weighting stack to +4.0; the aperture-as-input design absorbs the random-to-VMAT test shift; and the residual’s head-room is what multi-modal training converts into a further +2.8. The deployed model is a distilled student that is *faster and slightly better* than its teacher, evidence that at this task size the base-48 network is over-parameterized for the residual it must learn. **Limitations and generalizability.** All data come from one benchmark (abdomen/thorax, single machine model); cross-vendor and cross-site validation is future work. Ablation effects are single-fold ($n = 16$); the headline accuracy is the 5-fold CV. One geometrically hard thoracic case (large heterogeneous field) was the worst patient in all four challenge tasks simultaneously and passes clinically ($\gamma_{3/3} = 88.8$ – 97.4) while failing 1%/1 mm; the residual difficulty is geometric (steep dose–anatomy gradients), not modality-specific, and argues for cross-validated reporting (Fig. 5).

Future work. Within the present methodology, three directions follow directly from our measurements. (i) *DVH-aware selection*: our development was gamma-centric, and the DVH clinical score is the metric on which our submission ranks weakest relative to its gamma; internalizing the official DVH pipeline (structures from public segmentation models) for model selection is low-cost and likely worth more than further gamma tuning. (ii) *Student scaling and capacity*: base-32 matched and exceeded its teacher; mapping the accuracy–speed frontier over base-24/40 students would establish whether another $1.3\times$ is free. A complementary capacity study would overfit individual cases and identify the filter

Abdomen (1ABB006) — plan $\gamma 1\%/1\text{mm} = 96.6\%$



Lung (1THB002) — plan $\gamma 1\%/1\text{mm} = 98.0\%$



Hard case (1THB143) — plan $\gamma 1\%/1\text{mm} = 68.6\%$

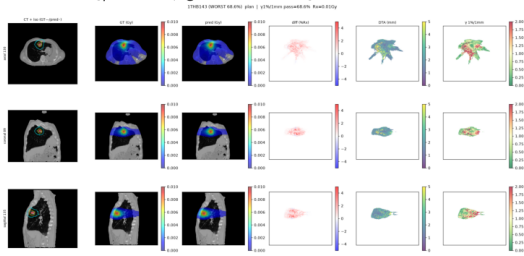


Fig. 5. Qualitative Photon-CT results on three cases spanning the difficulty range: abdomen (1ABB006, plan $\gamma 1/1 = 96.6$), lung (1THB002, 98.0), and the geometrically hard case 1THB143 (68.6; the worst patient in all four challenge tasks simultaneously, yet largely passing at 3%/3 mm: small-magnitude errors at steep lung–tissue interfaces). Columns: image with GT (solid) and predicted (dashed) isodose, MC dose, predicted dose, difference, distance-to-agreement, local $\gamma 1\%/1$ mm.

count at which single-case $\gamma 1/1$ saturates above 99%, demonstrating that the architecture is compact yet expressively sufficient. (iii) *Multi-modal training as the default*: the +2.8 cross-validated gain from joint real/synthetic-CT training held on every fold and should be the standard recipe wherever a paired MRI task exists.

Beyond the challenge, the engine’s speed and accuracy open clinical applications we intend to pursue. (iv) *Dose-aware image matching and adaptive triggering*: at ~ 0.1 s per control point, dose can be estimated on the image of the day (e.g. modern on-board CBCT) within the time of a normal image review.

Image match today is evaluated geometrically; even with the target well aligned, changes in surrounding tissue alter path lengths and distort the delivered distribution, and a real-time dose estimate would add the missing dosimetric component to the match decision and to adaptive-therapy triggering. (v) *Intrafraction dose reconstruction*: combining daily CBCT with low-dimensional deformation bases for soft-tissue motion and the recorded motion trace, the engine can reconstruct an estimated delivered dose per fraction, identifying systematic error and informing alignment and breathing-gate thresholds; a motion-phantom study along these lines is in progress at our institution. (vi) *Institutional beam-model tailoring*: refining the analytical prior and the residual network on commissioning data for our clinical beam model would tighten the physics prior further and is the natural path from challenge model to clinical decision support. (vii) *Gradient-based planning*: the engine is differentiable down to the fluence input, making it a natural physics layer for direct aperture/fluence optimization [8,9]; a proof-of-concept VMAT segment-weight optimization through the frozen engine is our next step.

5 Author Contributions (CReditT)

Kai Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Meixu Chen:** Software (deployment and runtime optimization), Investigation (acceleration study), Validation, Writing – review & editing. **Rui Yang:** Conceptualization, Methodology, Writing – review & editing.

6 Other Information

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